
How Reliable Are Agent Leaderboards? A Variance-Decomposition Analysis

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Abstract

1 Agent benchmark rankings drive deployment decisions, model release claims, and
2 policy citations, but their reliability has gone largely unmeasured. We apply a
3 generalizability theoretic framework to seven benchmarks on the Holistic Agent
4 Leaderboard (HAL), systematically decomposing score variance across model, task,
5 and scaffold, quantifying for the first time the effects of scaffold on benchmark
6 rank stability. We estimate the reliability of rankings at two levels of measurement:
7 the model and the model–scaffold pair, which illustrate divergent reliabilities on the
8 same data. Within a single benchmark, model-ranking reliability asymptotes near
9 $E\hat{\rho}^2 \approx 0.52$ under item-only scaling, well below conventional thresholds; broaden-
10 ing across the seven HAL benchmarks raises the projected ceiling to $E\hat{\rho}^2 \approx 0.88$.
11 The ceiling is therefore evaluation design-conditional: we show that noise from
12 model, scaffold, and benchmark interactions attenuates as scaffold and benchmark
13 conditions increase, not as item count increases, emphasizing that reliable model
14 rankings require diverse evaluation regimes rather than benchmark task additions.
15 We re-rank all seven HAL leaderboards and find substantial reordering— with mod-
16 els published as best dropping 16 ranks once scaffold and model contributions
17 are separated. The evidence from these analyses lead to actionable principles for
18 designing, reporting, and interpreting agent evaluations to improve the reliability
19 of AI-agent rankings: declare the object of measurement, attribute improvements
20 to the factor that varied, report reliability with uncertainty, and diversify contexts
21 rather than only task counts.

22 1 Introduction

23 Agent benchmarks are now the primary instrument for ranking frontier AI systems. Leaderboards
24 on popular benchmarks like SWE-bench [Jimenez et al., 2023], GAIA [Mialon et al., 2023], τ -
25 bench [Yao et al., 2024] are referenced in technical reports, system cards, and policy documents
26 Anthropic [2026], OpenAI [2026]. A model¹ that ranks first on one of these benchmarks is treated as
27 the state of the art for the (agentic) capability the benchmark purports to measure. Yet, the reliability
28 of these rankings has not been established.

29 This omission matters because reliability is not a property of a benchmark in the abstract; it depends
30 on *what the benchmark is being used to rank* and *over what scope*. A score reported as “Model X
31 achieves 47% on SWE-bench” under a single scaffold can be read as a claim about the base model or
32 about the specific (model, scaffold) system that was actually evaluated, and the statement does not
33 specify which. Most agent evaluation practice does not make this choice explicit, leaving consumers
34 without a declared object of measurement.

¹Throughout our analysis, “model” refers to a base language model paired with a specific reasoning-effort configuration; e.g., `gpt-5-high` and `gpt-5-minimal` are treated as distinct models.

This paper surfaces reliability in agent evaluations explicitly. Applying generalizability theory [Cronbach et al., 1972, Brennan, 2001] to seven agent benchmarks published on the Holistic Agent Leaderboard (HAL) [Kapoor et al., 2024], we decompose score variance into facets corresponding to model, item, scaffold, and their interactions, and compute reliability for two natural inferential targets: ranking models and ranking model–scaffold pairs. We find that the two yield divergent reliability profiles on the same data, that single-benchmark model rankings saturate well below conventional thresholds under item scaling alone while multi-benchmark aggregates do not, and that the resulting rank instability has direct consequences for the leaderboards practitioners use to select systems. The reliability of any such ranking is also conditional on the scaffold sample and the competitor set, since a tightly clustered frontier-model pool depresses between-model variance regardless of benchmark quality; we treat these as scope conditions on our results. We make four contributions:

- **Separating model and model–scaffold reliability across seven benchmarks.** We estimate ranking reliability for two objects of measurement on the same data—the model alone and the model–scaffold pair—and find that they yield substantially different and sometimes opposite reliability profiles: on AssistantBench, GAIA, and USACO, model–scaffold rankings are reliable while model rankings are not; on CoreBench-Hard, the reverse holds. Current reporting practice does not distinguish these objects, so a “reliable” benchmark for one inference can be unreliable for the other.
- **A design-conditional ceiling on model-ranking reliability.** Under item-only scaling within a single benchmark, we find that reliability asymptotes near $E\hat{\rho}^2 \approx 0.52$ while broadening across the seven HAL benchmarks raises the projected ceiling to ≈ 0.88 . The model main effect accounts for $\sim 3\%$ of total variance under the pooled decomposition, and the dominant components (benchmark $\times$ model, model $\times$ scaffold, benchmark $\times$ model $\times$ scaffold) attenuate with benchmark and scaffold count, not item count. In practice, reliable model rankings therefore require evaluation breadth, not more items.
- **A leaderboard case study showing that published rankings are often not statistically distinguishable.** On the HAL leaderboards, median inter-draw Spearman correlations of model capability estimates range from 0.16 to 0.47, and on five of seven benchmarks no pair of model ranks are mutually distinguishable at $\alpha = 0.1$. Re-ranking under the full decomposition reorders the leaderboards substantially: for example, the model published as best on τ -bench Airline drops 16 ranks, and the model published as best on SWE-bench Verified Mini drops to rank 12.
- **Principles for the design, reporting, and interpretation of agent evaluations.** We propose concrete changes addressed to benchmark designers, reporters of agent results, and consumers: declare the object of measurement, attribute improvements to the factor that varied, report reliability with uncertainty, and diversify contexts, not just task counts.

Section 2 reviews relevant work in measurement theory and agent evaluation. Section 3 defines the generalizability-theoretic setup and the two targets of comparison. Section 4 describes the dataset. Section 5 presents the empirical results: reliability under both units, variance decomposition, and the leaderboard case study. Section 6 develops the principles for more reliable agent evaluations. Section 6 discusses limitations and future work.

2 Related Work

Generalizability theory and measurement reliability. Generalizability theory (G-theory) is a measurement-theoretic framework for decomposing observed score variance into contributions from multiple facets and their interactions [Cronbach et al., 1972, Brennan, 2001, Cronbach and Shavelson, 2004, Shavelson et al., 1989, Revelle and Condon, 2019]. G-theory can be thought of as an extension of ANOVA: where ANOVA estimates how much variance is attributable to each factor, G-theory goes further by using those variance components to project reliability under hypothetical evaluation designs. Recent work has applied variance-decomposition methods to AI benchmarks more broadly, showing that design choices can induce substantial variability in rankings and conclusions [Hardy and Kim, 2026, Madaan et al., 2024, Hardy, 2025].

Agent benchmarks and their reliability. A growing ecosystem of benchmarks agent benchmark, software engineering [Jimenez et al., 2023], web navigation [Zhou et al., 2023, He et al., 2024], general-purpose assistance [Mialon et al., 2023], and customer-service Yao et al. [2024], use a leaderboard to indicate their best performing agent. These leaderboards rank using success rate

alone, making it difficult to infer possible inconsistencies in the agent’s performance. Several lines of work have introduced reliability metrics that surface sources of uncertainty in agent’s performance. Rabanser et al. [2026] and Yao et al. [2024] measure consistency by rerunning task evaluations. Others test robustness under varies conditions, prompt perturbations [Rabanser et al., 2026, Razavi et al., 2025], tool variation [Wang et al., 2026], and environmental errors injected during rollout [Rabanser et al., 2026, Kumar and Mishra, 2025]. This prior work asks whether a single agent performs consistently under reasonable variation in its task. We ask a different question: given an experimental design, how reliable are the comparative conclusions drawn from it? The answer depends not only on the agent under test but on the set of agents it is being compared against. To our knowledge, no prior work has studied the reliability of inferences drawn from agentic experiments.

3 Generalizability theory for agent benchmarks

G-theory can be used to explain the reliability of an experiment design by decomposing observed-score variability into variance components attributable to facets and their interactions. Reliability is defined as the fraction of observed-score variability attributable to the object of measurement rather than to sampling of other facets. In this section, we specify how we use G-theory to measure the reliability of claims made from agent evaluation data. First, we specify our facets, objects-of-measurement and formalizes a reliability metric (Section 3.1), then propose four variance decompositions (Section 3.2), and then specify our estimation procedure (Section 3.3).

3.1 Facets, objects of measurement, and universe scores

Applying G-theory to an agent benchmark requires three commitments: the facets, the population each facet is sampled from (its *universe*, in G-theory terminology), and the target of inference (the object of measurement). The same data can yield different reliability estimates under different choices, so we make these commitments explicitly.

Facets. A score is indexed by benchmark b , item i , model m , and agent scaffold a . Let $y_{b,i,m,a} \in \{0, 1\}$ be the observed binary score. We treat all four facets as random, with sampling populations defined below.

Sampling populations. We define a narrow universe tied to the design realized in HAL, and discuss broader extrapolations in Section 6.

- *Items* are treated as a random sample from the published item set of each benchmark. Generalization is to another draw from this set, not to the broader task domain the benchmark purports to represent (e.g., generalization is to “items like SWE-bench Verified Mini,” not to real-world software engineering). The construct-level question is a validity question, not a reliability question, as discussed in Section 6.
- *Scaffolds* are treated as a random sample from the set of scaffolds in current evaluation practice for each benchmark. Per-benchmark scaffold counts are 2–3, reflecting a field-wide pattern in which most agent benchmarks are run under one or two scaffolds and the universe of evaluated scaffolds is itself a selected sample (see Section 6). We pool across benchmarks in Decomposition 4 (Section 3.2) to recover statistical power.
- *Models* are treated as a random sample from frontier models evaluated on HAL (Section 4 during the study period. This universe is clustered: many models share base architectures and training pipelines. Clustering depresses σ_M^2 and is a feature of the current model landscape rather than an artifact of our measurement.

Object of measurement. Reliability is computed with respect to a chosen *object of measurement* o , the unit being ranked. We compute reliability for two objects, corresponding to two natural questions a leaderboard is used to answer: (i) ranking *models* ($o = m$), treating scaffold as a nuisance facet to be averaged over, and (ii) ranking *model–scaffold pairs* ($o = (m, a)$), treating the pair as the deployable unit and target of interest.

Relative reliability. Leaderboard use is a relative-decision problem: the question is which object ranks above which, not what each one’s absolute score is. We therefore report relative reliability throughout, denoted $E\rho^2$, in which nuisance-facet main effects shared across all objects drop out and

only object-by-facet interactions and residual terms enter the error. For an object o aggregated over n_i items and n_a scaffolds,

$$E\rho^2 = \frac{\sigma_o^2}{\sigma_o^2 + \sigma_{\text{rel-error}}^2(n_i, n_a, \dots)}, \quad (1)$$

where $\sigma_{\text{rel-error}}^2$ collects object-by-facet interaction variances and the residual, each scaled by the relevant sample sizes; main effects of i , a , and b alone do not appear in the denominator. The function $\sigma_{\text{rel-error}}^2(\cdot)$ is the D-study (decision-study) projection: it describes how error shrinks as the number of sampled items, scaffolds, or benchmarks grows. The full expression for each decomposition is given in Appendix C.7. The framework extends straightforwardly to absolute reliability (Φ) for threshold decisions, e.g., certifying that a model exceeds a fixed performance bar [Brennan, 2003].

3.2 Variance decomposition models

We estimate four variance decomposition models of increasing generalizability: three per-benchmark (Decomposition 1–3) and one pooled ecosystem model (Decomposition 4). Code and details for each are found in Appendix C.

Notation. Let $b \in \mathcal{B}$ index benchmark, $i \in \mathcal{I}_b$ index item (task), $m \in \mathcal{M}_b$ index model, and $a \in \mathcal{A}_b$ index agent scaffold. Let $y_{b,i,m,a}$ be the binary score when observed. The canonical GLMM is $y_{b,i,m,a} \sim \text{Bernoulli}(p_{b,i,m,a})$, $\text{logit}(p_{b,i,m,a}) = \eta_{b,i,m,a}$, with η expressed as a sum of random effects corresponding to facets and interactions (below). For LME fits, we treat y as approximately continuous and use an identity link; for GLMM and Bayesian GLMM, we use the logit link.

Decomposition 1 (naïve per-benchmark; object is model) This decomposition mirrors a common leaderboard assumption: scores vary by model and by item, with all other variability absorbed into residual noise. For each benchmark b : $\text{logit}(p_{i,m}) = \beta_0 + u_i^{(I)} + u_m^{(M)}$, with $u_i^{(I)} \sim \mathcal{N}(0, \sigma_I^2)$ and $u_m^{(M)} \sim \mathcal{N}(0, \sigma_M^2)$.

Object of measurement: model m . *Purpose:* quantify the optimism induced by omitting agent scaffold variance and model–agent interactions.

Decomposition 2 (per-benchmark; object is model–agent pair) This model redefines the object as the *model–agent pair*, absorbing agent and model $\times$ agent variability into “signal”: $\text{logit}(p_{i,(m,a)}) = \beta_0 + u_i^{(I)} + u_{m:a}^{(MA)}$, where $u_{m:a}^{(MA)} \sim \mathcal{N}(0, \sigma_{MA}^2)$.

Object of measurement: pair (m, a) . *Interpretation:* reliability for ranking *deployable systems*, not bare models.

Decomposition 3 (per-benchmark; separates model, agent, and interactions) Model 3 is the per-benchmark design intended to reflect agentic evaluation reality: scores depend on model, item, agent harness, and their interactions:

$$\eta_{i,m,a} = \beta_0 + u_i^{(I)} + u_m^{(M)} + u_a^{(A)} + u_{i:m}^{(IM)} + u_{i:a}^{(IA)} + u_{m:a}^{(MA)}. \quad (2)$$

We treat the 3-way interaction $u_{i:m:a}^{(IMA)}$ as *confounded with residual* because several benchmarks lack repeated runs per (i, m, a) , making separate identification unstable.

Objects of measurement: model m (primary) and, secondarily, pair (m, a) . *Use:* compute D-study reliability under hypothetical changes in the number of items and/or agent scaffold, explicitly accounting for model–agent coupling.

Decomposition 4 (pooled ecosystem model across benchmarks) To use shared signal across harnesses and benchmarks (notably via HAL-generalist), we pool all benchmarks and include benchmark-level facets and cross-benchmark interactions:

$$\eta_{b,i,m,a} = \beta_0 + u_b^{(B)} + u_{b:i}^{(BI)} + u_m^{(M)} + u_a^{(A)} + u_{b:i:m}^{(BIM)} + u_{b:i:a}^{(BIA)} + u_{m:a}^{(MA)} + u_{b:i:m:a}^{(BIMA)} + u_{b:m}^{(BM)} + u_{b:a}^{(BA)} + u_{b:m:a}^{(BMA)}. \quad (3)$$

All random effects are modeled as mean-zero Gaussian with component variances to be estimated. This model supports the central question: *even if we scale item counts, is model-ranking reliability fundamentally bounded by cross-benchmark and harness variability?*

Table 1: Benchmark design summary. Counts denote unique levels of each facet.

Benchmark	models	Tasks	Agents	model–Scaffold Pairs
AssistantBench	18	33	2	20
CoreBench-Hard	32	45	2	34
GAIA	21	165	2	23
SciCode	17	65	3	20
SWE-bench (Mini)	24	50	2	26
TAU-bench (Airline)	23	50	3	26
USACO	13	307	2	15

3.3 Estimation approaches

We fit Bayesian logistic mixed models with weakly informative priors on fixed effects and half-Student- t (or equivalent) priors on standard deviations: $\sigma_k \sim \text{Half-}t(\nu, 0, s)$, with conservative (ν, s) to regularize variance components away from degenerate solutions without dominating the likelihood. Posterior draws are used to compute posterior distributions over variance shares and reliabilities. All parameters across all models satisfy $\hat{R} < 1.05$. For estimated capability score for posterior ranking, Best Linear Unbiased Predictors (BLUPs) were extracted as draws of the estimated random effects. Each decomposition (1-4) is fit using three additional estimation methods to stress-test conclusions under small- n and non-Gaussianity. For Decomposition 4, we detail its identifiability in Appendix C.10 and demonstrate the stability global findings via four estimation method comparisons, all of which are detailed in Section C and Section E.

4 Data

We analyze agent rollouts from seven benchmarks distributed via HAL Kapoor et al. [2024]: AssistantBench [Yoran et al., 2024], CoreBench Hard [Siegel et al., 2024], GAIA [Mialon et al., 2023], SciCode [Tian et al., 2024], SWE-bench Verified Mini [Jimenez et al., 2023], τ -bench Airline [Yao et al., 2024], and USACO [Shi et al., 2024], covering web navigation, scientific reproducibility, multi-step assistance, scientific programming, software engineering, customer-service interaction, and competitive programming respectively (full descriptions in Appendix A).

Our analysis covers 20,144 total agent rollouts. Each rollout is over a single task, model, reasoning-effort, and scaffold. Because our analysis is mostly interesting the relationship between the model-variant, scaffold and benchmarks, we join the reasoning-effort with the model to be our model-facet. For our analysis, each time model is stated it refers to the model at a specific-reasoning effort (i.e. ‘gpt5-high’ and ‘gpt5-minimal’ are separated). Task counts range from 33 (AssistantBench) to 307 (USACO), model-Variant counts from 10 to 23, and each benchmark has 2 or 3 distinct scaffolds. All outcomes are task-level binary scores ($\text{score}_{ijk} \in \{0, 1\}$). The following benchmarks use a subset of tasks from the full benchmark, based on the available on the HAL leaderboard Kapoor et al. [2025]: AssistantBench, GAIA, Scicode and SWE-bench Verified. Table 1 summarizes the design.

4.1 Data Limitations and Identifiability

Notice that every model–scaffold–benchmark combination is observed; the full coverage matrix is in Figure 7. Several design features merit emphasis. First, the number of agent scaffolds per benchmark is extremely small (2–3), making scaffold variance difficult to estimate yet potentially consequential. Second, one scaffold—HAL-GENERALIST—appears in every benchmark, providing the critical thread of connectivity across the ecosystem and making it the linchpin of the pooled analysis used in Section 5.2-5.3 by anchoring model comparisons across otherwise disjoint evaluation pipelines and preventing global model variance from being an artifact of pooling. Care is taken to define Decomposition 4 that can be identified given the data. The design needs enough connectivity to separate “model signal” from “non-vanishing ecosystem heterogeneity,” which HAL-GENERALIST provides. The pooled model borrows strength only because real overlaps exist (shared harness and shared models/items within benchmarks), so the decomposition is data-supported rather than purely model-imposed.

Furthermore, the findings are estimator- and method-robust. The same qualitative conclusions—nontrivial variance outside the model main effect implying limited model-ranking reliability—appears under Bayesian GLMM, frequentist GLMM/LME baselines, and nonparametric distance components, reducing concern about parametric or hierarchical artifacts. We report details for identifiability in Section C.10 and additional estimation method information in §C and E.

5 Results

We present the empirical results in three parts. Section 5.1 estimates reliability separately for each of the seven benchmarks under two units of comparison: the model alone and the model–scaffold pair. Section 5.2 pools across benchmarks to estimate a structural reliability ceiling for model ranking, and decomposes total score variance to identify the components that determine it. Section 5.3 translates the estimated variance components into uncertainty on individual model scores and rank positions on the HAL leaderboards.

5.1 Reliability depends on the unit of comparison

In this section, we show how reliability can change depending on which object of measurement is selected. A practitioner may ask, *which model performs the best for X benchmark?* or *which model–scaffold pair performs the best for X benchmark?*, and these may lead to different answers on the same dataset. Each of these can lead to a different conclusion being from the same dataset. Corresponding to these two questions, Figure 1 shows posterior reliability estimates with 95% confidence intervals for each benchmark under two units of comparison: the model alone (Decomposition 3), where scaffold is treated as a nuisance facet, and the model–scaffold pair (Decomposition 2 or 3), where scaffold is absorbed into the object of measurement.

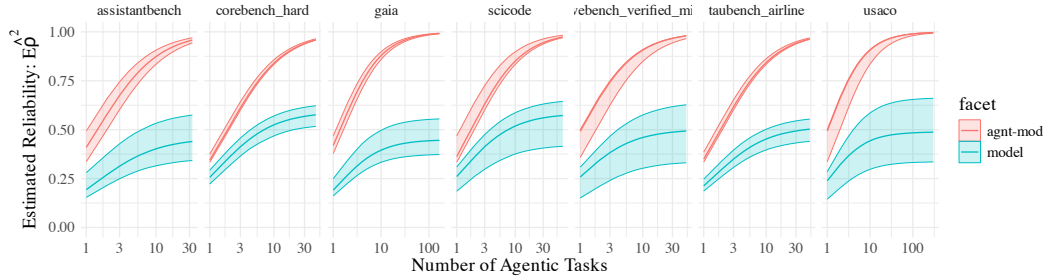


Figure 1: Estimated Reliability by number of items used, comparing LLM and LLM + scaffold as different objects of measurement. D-study estimates are sampled directly from the posteriors of the per-benchmark estimations of Eq. 2. For each point $\hat{R} \leq 1.05$. Credible intervals are 95% HDI, and effect estimates are medians to account for skewness in posterior. Variance components aggregated after inverse logit transform, thus reliabilities represent the observation space.

Reliability varies substantially across benchmarks, and which unit yields higher reliability is itself benchmark-specific. Several benchmarks — AssistantBench, GAIA, and USACO — show pair-unit reliability approaching 1.0 while model-unit reliability remains below 0.5, indicating that inferences about model performance alone are weak, whereas inferences about model–scaffold performance can be strong. Moreover, this highlights several cases where making model-level inferences, without factoring the agent contribution, may lead up unreliable inferences. The naïve decomposition implicit in current reporting practice overstates model-unit reliability relative to the full decomposition (Appendix C and D).

5.2 Item-only scaling saturates to ceiling; benchmark diversity raises the ceiling

Figure 2 shows that the reliability bottleneck is not simply the number of tasks, but the diversity of contexts over which those tasks are sampled. When all additional tasks are drawn from a single benchmark, reliability improves at first but plateaus around $E\hat{\rho}^2 \approx 0.52$. This is an item-only ceiling: variance from model–benchmark and model–scaffold interactions remain even as item-level noise vanishes. By contrast, when the same evaluation budget is spread across benchmarks,

benchmark-conditioned model heterogeneity is also averaged down, and the projected reliability rises substantially, approaching $E\hat{\rho}^2 \approx 0.88$ for the seven-benchmark panel. This shows that scaling an agent leaderboard by adding more tasks from the same benchmark is a weak remedy; the effective design intervention is to broaden the evaluation across independently informative benchmark and scaffold contexts. The behavior follows from the closed-form D-study projection for model-ranking reliability under Decomposition 4:

$$E\rho_M^2(n_b, n_i, n_a) = \frac{\sigma_M^2}{\sigma_M^2 + \frac{\sigma_{BM}^2}{n_b} + \frac{\sigma_{MA}^2}{n_a} + \frac{\sigma_{BMA}^2}{n_b n_a} + \frac{\sigma_{BIM}^2}{n_b n_i} + \frac{\sigma_{BIMA,e}^2}{n_b n_i n_a}}. \quad (4)$$

Item-indexed components attenuate with n_i ; benchmark-indexed components attenuate with n_b ; scaffold-indexed components attenuate with n_a . Holding n_a fixed,

$$\lim_{n_i \rightarrow \infty} E\rho_M^2(n_b, n_i, n_a) = \frac{\sigma_M^2}{\sigma_M^2 + \frac{\sigma_{BM}^2}{n_b} + \frac{\sigma_{MA}^2}{n_a} + \frac{\sigma_{BMA}^2}{n_b n_a}}. \quad (5)$$

The ceiling is design-conditional, not universal. At $n_b = 1$, benchmark–model heterogeneity does not average out, and the asymptote sits where Figure 2 places it; at $n_b = 7$, the same heterogeneity is divided by seven.

Figure 3 shows why the single-benchmark plateau is so low: The model main effect accounts for $\sim 3\%$ of total variance under Decomposition 4 (2–14% per-benchmark under Decomposition 3, see Appendix D and 7), smaller than item-conditioned and scaffold-related components. On a single benchmark, score differences between models are dominated by the item and scaffold draw and a single-benchmark leader position is not statistically separable from the field (see Section 5.3).

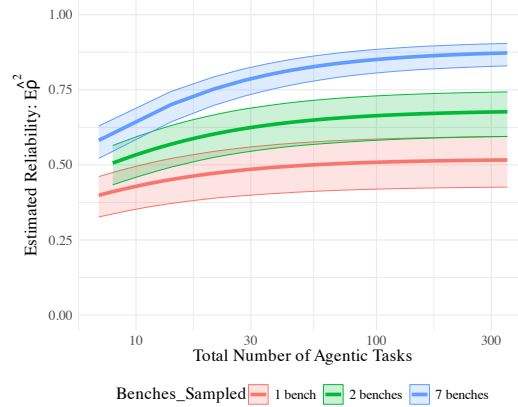


Figure 2: Full variance decomposition model for LLM rankings on AI-agent leaderboards. D-study estimates are sampled directly from the posteriors of the per-benchmark estimations of Eq. 3. Scenarios estimated include sampling 350 tasks from one benchmark, 175 tasks from two benchmarks each, or 50 from 7 each. For each point $\hat{R} \leq 1.05$. Credible intervals are 95% HDI, and effect estimates are medians to account for skewness in posterior. Variance components aggregated after inverse logit transform, thus reliabilities represent the observation space.

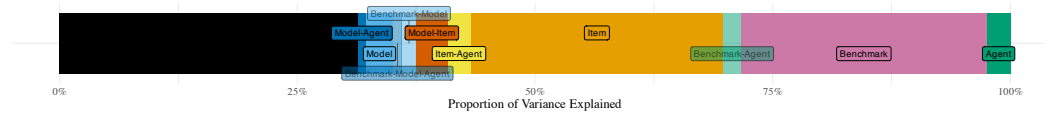


Figure 3: Full variance decomposition based on Eq. 3 with proportions estimated with Eq. 11. Variance decomposition for individual benchmarks is shown in Fig. D.

5.3 Consequences for current agent leaderboards

In this section, we demonstrate that reliability bounds have direct consequences for the rankings published on popular leaderboards, using the HAL leaderboards as illustrative examples². Across the full posterior, median rank stability measured by median pairwise inter-draw Spearman correlations of Best Linear Unbiased Predictors (BLUPs) was low, shown in Table 2. Therefore, published ordering reflects a particular choice of prompts and scaffolds, and a different draw could plausibly invert it.

Table 2: Rank Uncertainty: Median Rank Correlation across BLUP Posterior Draws

Assistantbench	Corebench	GAIA	Scicode	SWEbench verified	τ -bench airline	USACO
0.164	0.219	0.26	0.265	0.47	0.186	0.374

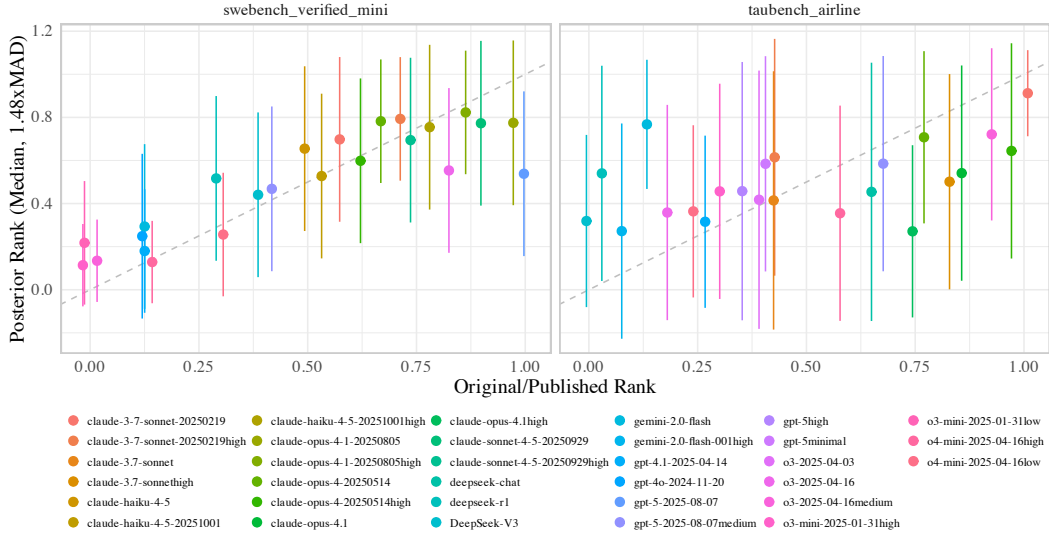


Figure 4: Published rankings and draw-wise Eq. 3 posterior median BLUP rankings of BLUPs with $1.48 \times \text{MAD}$. Estimates for all benchmarks in Figure 11.

The same logic applied to the published leader gives a worst-case rank drop. Taking the lower edge of the leader’s 95% confidence interval and asking how many models have point estimates above that lower edge yields the rank position the leader could plausibly occupy under a less favorable draw of items and scaffolds.

Figures 5 and 12 report the full ranking shifts for each benchmark. The model ranked first under Decomposition 1 falls to rank 12 on SWE Bench under the more complete Decomposition 4; the largest single rank shift is 16 positions with τ -bench. In fact, few benchmark capability estimates are mutually distinguishable across draws ($\alpha = 0.1$), with only GAIA and SWE Bench producing any statistically distinguishable scores. These are not adversarial constructions but direct readings of the differences of random intercepts and standard errors the variance decomposition produces.

Rankings instability, which is expectedly more pronounced than continuous score variability, is also visible when we re-rank under a more complete variance decomposition. Figure 5 shows model rankings on τ -bench Airline and SWE-Bench Verified (Mini) under the naive decomposition (Decomposition 1, the implicit assumption of current leaderboard practice, left) and then under the pooled decomposition (Decomposition 4, which separates scaffold and model–scaffold variance from model variance, right). The Decomposition 1 ranking (the naive approach currently used in the field) conflates model capability with model–scaffold compatibility: a model paired with a scaffold that suits it well looks better than its underlying capability warrants, and a model paired with a less suitable scaffold looks worse. Decomposition 4 separates the two contributions, and the reordering shows which models were over- or under-credited by the scaffolds they were tested with.

²Using 1000 posterior draws from Decomposition 4.

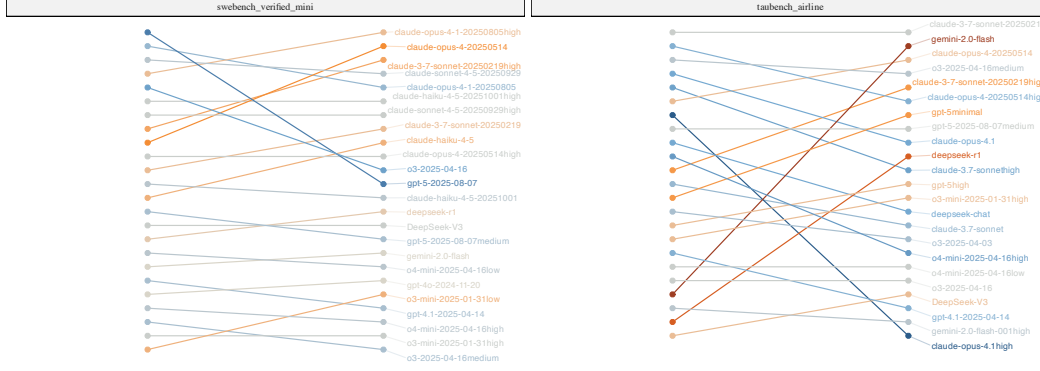


Figure 5: τ bench-Airline and SWE Bench Verified (Mini) rankings based on current practice (left) compared to Best Linear Unbiased Predictors extracted from the posterior of Eq. 3 (right). The other benchmarks of this study can be found in Fig. 12.

6 Principles for Reliable Agent Evaluation

Four principles follow from the findings in Section 5, addressed to benchmark designers, reporters, and consumers of agent evaluations.

Declare the object of measurement. Every score should declare what it ranks: the model, the model–scaffold pair, or some other unit. The choice determines which components are signal (σ_M^2 vs. σ_{MA}^2 absorbed into signal) and which are error, and yields divergent reliability on the same data (Section 5.1). For example, “47% on SWE-bench” under a single scaffold is a result about the (model, scaffold) system, not the model.

Attribute improvements to the factor that varied. Model releases often change base model, scaffold, tools, prompting, and inference budget at once. Aggregate gains cannot be decomposed post hoc: a model-level claim requires multiple scaffolds, a scaffold-level claim multiple base models. Section 5.3 showed the model published as best on τ -bench Airline drops 16 ranks once contributions are separated.

Report reliability with uncertainty. A score should come with a reliability estimate against the declared object and a credible interval. The minimum reportable set consists of the point estimate, credible interval, object of measurement, and estimation method. Without these, the leaderboard consequences in Section 5.3 are invisible to consumers.

Diversify contexts, not just task counts. Section 5.2 shows that single-benchmark model-ranking reliability saturates near $E\hat{\rho}^2 \approx 0.52$ under item-only scaling, because σ_{BM}^2 , σ_{BMA}^2 , and σ_{MA}^2 do not attenuate with item count. Broadening across benchmarks attenuates the benchmark-conditioned terms and raises the projected ceiling to ≈ 0.88 at $n_b = 7$ in our considered benchmark set. The corresponding design prescription has three parts. First, allocate evaluation budget across independently informative benchmarks rather than concentrating it within one. Second, sample multiple scaffolds per benchmark; current per-benchmark counts ($n_a \in \{2, 3\}$) leave σ_{MA}^2 poorly estimated and absorbed into non-attenuating error. Third, report a D-study curve over n_i , n_a , and n_b so consumers can pick a design that meets a target reliability at minimum cost. The diversification budget should be set in proportion to the inference the leaderboard is intended to support: rankings of deployable model–scaffold systems require fewer scaffold draws than rankings of base-model capability across the task space.

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763 be described.

764 Appendix

765 Limitations and Future Work

766 **Reliability is necessary but not sufficient for valid evaluation.** The bounds we estimate describe
767 how consistently a benchmark differentiates the objects it ranks, not whether the rankings correspond
768 to underlying capability or to any external ground truth: a benchmark can be reliable but invalid.
769 The object-of-measurement problem we document is in this sense a construct validity problem in
770 disguise, since reliability against an undefined construct cannot be assessed in principle [Cronbach
771 and Meehl, 1955, Salaudeen et al., 2025]. Our bounds are also within-benchmark: whether any object
772 of measurement supports generalization *across* benchmarks, e.g., whether a model that ranks first
773 on SWE-bench Verified Mini also ranks first on τ -bench Airline, is a separate validity question our
774 framework does not directly answer. Future work should complement the reliability framework with
775 validity studies, including external-criterion validation and cross-benchmark transfer analyses.

776 **Scope conditions on the headline ceiling.** The reliability bounds we report are joint properties of
777 the benchmark, the scaffold sample, the item sample, and the competitor set being ranked. A clustered
778 frontier-model pool depresses σ_M^2 regardless of benchmark quality, lowering reliability not because
779 the benchmark is poorly designed but because it is being used to distinguish systems too close for
780 its resolution; hence, the ceiling we find should be read as a statement about current leaderboards
781 for the current frontier-model competitor set. Scaffolds are selected artifacts, often co-developed
782 with the benchmarks they evaluate, and per-benchmark counts ($n_a \in \{2, 3\}$) reflect standard practice
783 across the field rather than a feature of HAL; pooling across benchmarks in Decomposition 4 partially
784 addresses but per-benchmark scaffold-variance claims remain weaker, and pooling itself depends on
785 HAL-GENERALIST as the cross-benchmark anchor (Appendix C.10). Future work should construct
786 evaluation datasets that systematically span larger scaffold libraries, and should track how the ceiling
787 shifts as the model pool evolves.

788 **Latent-scale reliability.** We estimate reliability on the latent logit scale, while leaderboards report
 789 observed proportions. The two scales do not translate one-to-one, so our numbers should be read as
 790 approximate bounds on observed-scale rank stability. Future work could entail a sensitivity analysis
 791 comparing the scales.

792 **Raising reliability bounds.** The empirical scope of this work is the characterization of reliability
 793 bounds rather than their displacement. Validating interventions that raise the ceiling is the subject of
 794 subsequent work, since each candidate intervention introduces its own measurement-design tradeoffs.
 795 Candidate interventions include item selection by item discrimination, scaffold sampling under a
 796 defined scaffold family, and item-quality auditing; empirical evaluation of these, alongside methods
 797 that reduce the measurement cost of the recommendations in Section 6, are the most direct extensions.

798 A Benchmark Descriptions

799 **AssistantBench** Tasks consist of time-consuming, busy-work tasks that an average person may
 800 face, seeking information from the web.

Which gyms near Tompkins Square Park (<200m) have fitness classes before 7am?

801

802 **CORE-Bench Hard** Tasks ask an agent to computationally reproduce specific quantitative results
 803 from a published scientific paper, given a code repository, dataset, and research paper.

Run the main.py file three times. First, with `config/uci.json`, the preprocessing task, and the CTGCN-C method. Second, with `config/uci.json`, the embedding task, and the CTGCN-C method. Third, using Python3 with `config/uci.json` and the link-pred task.

804

805 **GAIA** GAIA consists of tasks requiring multi-step tool use — including web browsing, code
 806 execution, file reading (PDFs, spreadsheets, audio), and multimodal understanding.

(Given an Excel file) The attached Excel file contains the sales of menu items for a local fast-food chain. What were the total sales that the chain made from food (not including drinks)? Express your answer in USD with two decimal places.

807

808 **SciCode** Tasks consist of research-level coding problems decomposed into subproblems drawn
 809 from actual scientific work across physics, chemistry, biology, math, and materials science.

Main problem: Reproduce the Chern number phase diagram of the Haldane model on a hexagonal lattice. Subproblem 1.1: Write a Haldane model Hamiltonian on a hexagonal lattice, given: wavevector components k_x and k_y , lattice spacing a , nearest-neighbor coupling constant t_1 , next-nearest-neighbor coupling constant t_2 , phase φ for next-nearest-neighbor hopping, and on-site energy m . Output: a 2×2 matrix. Subproblem 1.2: Calculate the Chern number using the Haldane Hamiltonian, given the grid size δ for discretizing the Brillouin zone...

810

811 **SWE-bench Verified Mini** A subset of SWE-bench Verified where each task gives the agent a real
 812 GitHub issue description and a full Python repository, and requires the agent to produce a code patch
 813 that makes failing unit tests pass.

(Given a repo and issue description) Navigate the repository, identify the relevant code path in `django/db/models/query.py`, and produce a `.patch` file that makes the FAIL_TO_PASS unit tests pass without breaking existing PASS_TO_PASS tests.

814

815 **τ -bench Airline** Tasks simulate a realistic airline customer service scenario in which a human user
 816 (played by another LLM) contacts an agent with requests like rebooking a flight, adding a passenger,
 817 or canceling a reservation. The agent must follow a detailed airline policy document and a limited set
 818 of callable functions.

(Given the airline’s policy and pre-defined functions) Your user id is *daiki_muller_1116*. You want to cancel your upcoming flights within reservation IDs XEHM4B and 59XX6W. If the agent says either reservation has basic economy flights, ask to upgrade them to economy first and then cancel them. You are very persistent and terse but clear. After the third agent message, also ask whether you have any other upcoming flights and what their total cost is.

USACO Tasks are problems from the USA Computing Olympiad, spanning four difficulty tiers (Bronze through Platinum).

Farmer John has N cows ($2 \leq N \leq 10^5$), each liking exactly one type of hay h_i . He can host focus groups over contiguous ranges of cows — if more than half the cows in a group prefer the same type, all cows switch to that type. He wants to know which types of hay can become universally liked. Given T test cases, each with N and a list of h_i values, output all achievable universal hay types in increasing order, or -1 .

B Coverage Matrix

Below is the list of 2-3 agents used to test each of the benchmarks.

Benchmark	Agent Names
assistantbench	hal_generalist, assistantbench_browser_agent
corebench-hard	hal_generalist, coreagent
taubench-airline	hal_generalist, taubench_fewshot, taubench_tool_calling
swebench-verified-mini	hal_generalist, sweagent
scicode	hal_generalist, tool_calling_agent, scicode_zero
gaia	hal_generalist, hf_open_deep_research
usaco	hal_generalist, usaco_episodic_semantic

Table 3: Trace summary by benchmark.

Each model was not uniformly tested with each benchmark and agent. As show in Figure 6, about half the agents have only been tested on a single benchmark. Similarly, many model (11/30) are tested with a single agent.

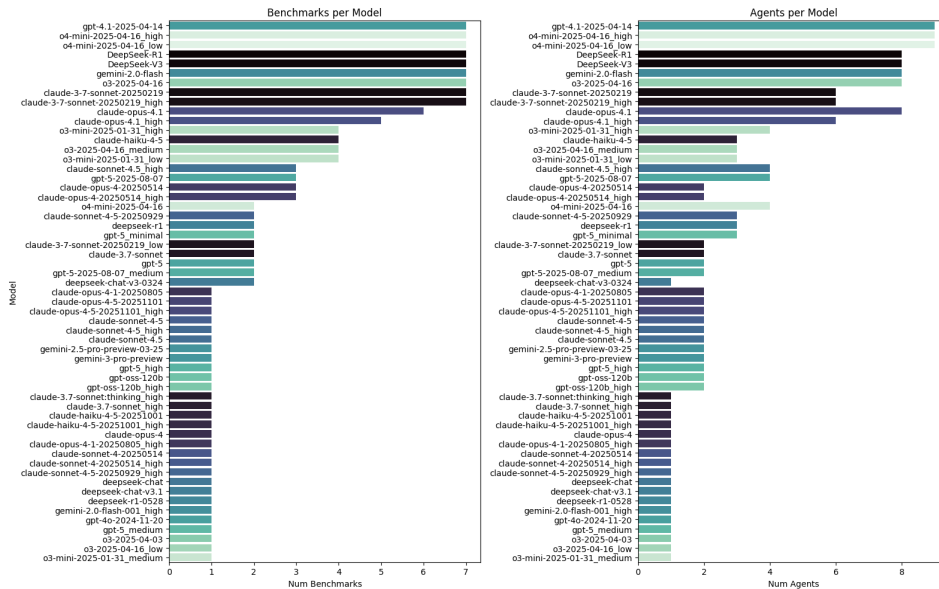


Figure 6: Number of benchmarks tested (left) and agents tested (right) per model.

The full breakdown of which models, agents and benchmarks have been tested can be seen in Figure 7.

C Methods Details

C.1 Model Specifications in Detail

We provide the full random-effects structure for each model using standard mixed-model notation. All models assume Bernoulli outcomes with logit link for Methods 2–4, and Gaussian identity link for Decomposition 1. We denote indices as: i for item (task), j for model (model name $\times$ reasoning-effort), k for agent scaffold, and b for benchmark.

C.1.1 Decomposition 1: Naïve

$$\text{logit } \Pr(Y_{ij} = 1) = \mu + t_i + m_j, \quad t_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_t^2), \quad m_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_m^2). \quad (6)$$

Fit per benchmark. The residual on the logit scale is the standard logistic variance $\pi^2/3 \approx 3.29$.

brms specification:

```
brm(score ~ 1 + (1|task_id) + (1|model),
    family = bernoulli("logit"), ...)
```

C.1.2 Decomposition 2: Model–Scaffold Pair

$$\text{logit } \Pr(Y_{i,jk} = 1) = \mu + t_i + p_{jk}, \quad p_{jk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_p^2). \quad (7)$$

Fit per benchmark. The model $\times$ agent interaction is absorbed into the pair term p_{jk} .

brms specification:

```
brm(score ~ 1 + (1|task_id) + (1|model:agent_name),
    family = bernoulli("logit"), ...)
```

C.1.3 Decomposition 3: Full Per-Benchmark

$$\begin{aligned} \text{logit } \Pr(Y_{ijk} = 1) &= \mu + t_i + m_j + a_k + (tm)_{ij} + (ta)_{ik} + (ma)_{jk}, \\ t_i &\sim \mathcal{N}(0, \sigma_t^2), \quad m_j \sim \mathcal{N}(0, \sigma_m^2), \quad a_k \sim \mathcal{N}(0, \sigma_a^2), \\ (tm)_{ij} &\sim \mathcal{N}(0, \sigma_{tm}^2), \quad (ta)_{ik} \sim \mathcal{N}(0, \sigma_{ta}^2), \quad (ma)_{jk} \sim \mathcal{N}(0, \sigma_{ma}^2). \end{aligned} \quad (8)$$

The three-way interaction $(tma)_{ijk}$ is confounded with the residual in the absence of within-cell replication.

brms specification:

```
brm(score ~ 1 + (1|task_id) + (1|model_name) + (1|agent_name)
    + (1|task_id:model_name) + (1|task_id:agent_name)
    + (1|model_name:agent_name),
    family = bernoulli("logit"), ...)
```

C.1.4 Decomposition 4: Pooled Leaderboard

$$\begin{aligned} \text{logit } \Pr(Y_{bijkl} = 1) &= \mu + b_b + (bt)_{bi} + m_j + a_k \\ &\quad + (btm)_{bij} + (bta)_{bik} + (ma)_{jk} + (btma)_{bjk} \\ &\quad + (bm)_{bj} + (ba)_{bk} + (bma)_{bjk}. \end{aligned} \quad (9)$$

All random effects are independent and normally distributed. Items are nested within benchmarks, so t_i does not appear as a main effect; its role is absorbed by $(bt)_{bi}$.

brms specification:

```

857 brm(score ~ 1 + (1|benchmark) + (1|benchmark:task_id)
858     + (1|model_name) + (1|agent_name)
859     + (1|benchmark:task_id:model_name)
860     + (1|benchmark:task_id:agent_name)
861     + (1|model_name:agent_name)
862     + (1|benchmark:task_id:model_name:agent_name)
863     + (1|benchmark:model_name)
864     + (1|benchmark:agent_name)
865     + (1|benchmark:model_name:agent_name),
866     family = bernoulli("logit"), ...)

```

867 C.1.5 Decomposition 4 Multivariate Extension

868 The multivariate extension models two outcomes simultaneously:

$$\begin{pmatrix} \text{logit Pr}(\text{score}_{bijkl} = 1) \\ \log(\text{tokens}_{bijkl}) \end{pmatrix} = \boldsymbol{\mu} + [\text{same random-effects structure as Model 4}], \quad (10)$$

869 with outcome-specific variance components and a residual correlation parameter ρ_ε between score
870 and log-tokens.

871 brms specification:

```

872 brm(bf(mvbind(score, log(total_tokens)) ~ 1
873     + (1|benchmark) + (1|benchmark:task_id)
874     + (1|model_name) + (1|agent_name)
875     + (1|benchmark:task_id:model_name)
876     + (1|benchmark:task_id:agent_name)
877     + (1|model_name:agent_name)
878     + (1|benchmark:task_id:model_name:agent_name)
879     + (1|benchmark:model_name)
880     + (1|benchmark:agent_name)
881     + (1|benchmark:model_name:agent_name)), ...)

```

882 C.2 Bayesian Estimation Details

883 **Priors.** All models use brms default weakly informative priors: half-Student- $t(3, 0, 2.5)$ on random-
884 effect standard deviations and a Student- $t(3, 0, 2.5)$ prior on the intercept. These priors regularize
885 estimates away from boundary values ($\sigma = 0$) while remaining sufficiently diffuse to let the data
886 dominate.

887 **Sampling.** Models 1–3 use 4 chains $\times$ 1,000 iterations with a thinning factor of 2, yielding 2,000
888 posterior draws per model. Model 4 uses 4 chains $\times$ 2,000 iterations with thinning of 5, yielding
889 1,600 draws. The target acceptance rate is set to `adapt_delta = 0.95` across all models to reduce
890 divergent transitions in the complex random-effects structure.

891 **Convergence.** All parameters across all models achieve $\hat{R} < 1.05$, the standard threshold for
892 adequate mixing. We additionally inspect trace plots and effective sample sizes; the latter exceed 400
893 for all variance components.

894 **Variance proportion computation.** From each posterior draw $s = 1, \dots, S$, we compute the
895 variance share for facet f as

$$\pi_f^{(s)} = \frac{(\sigma_f^{(s)})^2}{\sum_g (\sigma_g^{(s)})^2 + \pi^2/3}, \quad (11)$$

896 where the sum runs over all random-effect standard deviations and $\pi^2/3$ is the logistic residual
897 variance. Posterior summaries (mean, median, HDI) are computed over the $\{\pi_f^{(s)}\}$ draws.

898 C.3 Distance Components (DISCO)

899 The DISCO decomposition [Rizzo and Székely, 2010] generalizes classical ANOVA to arbitrary
900 metric spaces. For K groups with n_k observations each, define the within-group dispersion as

$$\mathcal{S}_W = \sum_{k=1}^K \frac{1}{2n_k} \sum_{i < j} \|X_{ki} - X_{kj}\|, \quad (12)$$

901 and the total dispersion as

$$\mathcal{S}_T = \frac{1}{2N} \sum_{i < j} \|X_i - X_j\|, \quad (13)$$

902 where $N = \sum_k n_k$. The between-group component is $\mathcal{S}_B = \mathcal{S}_T - \mathcal{S}_W$, and the proportion attributable
903 to the grouping factor is $\mathcal{S}_B/\mathcal{S}_T$.

904 For multi-facet designs, we compute DISCO sequentially for each facet, using the dispersion ratio
905 as the analog of η^2 . Because DISCO does not produce a residual term, the facet proportions do
906 not generally sum to one when computed independently; we normalize to aid comparison with the
907 parametric estimates.

908 *Remark C.1.* DISCO attributes substantially more dispersion to interaction terms than parametric
909 methods. This occurs because pairwise distances capture nonlinear dependencies—such as a model
910 performing anomalously well on a specific item cluster—that additive random-effects models cannot
911 represent. The DISCO interaction estimates should therefore be interpreted as an upper bound on the
912 importance of non-additive effects, complementing the more conservative parametric decomposition.

913 C.4 Details On Compute-Usage

914 Estimation of all four methods across all four decompositions took 12 total hours on an Apple M1
915 Max.

916 C.5 Estimation Methods

917 Each of the four models is estimated by four methods, chosen to span the space from conventional to
918 robust. In the main body, we report Method C. Findings are robust to estimation method.

919 C.5.1 Method A: Linear Mixed Effects (LME)

920 We fit a Gaussian identity-link model via restricted maximum likelihood (REML). Although the
921 binary outcome violates normality, LME provides a familiar baseline and is the most commonly used
922 variance-component estimator in G-theory applications. Variance components are extracted directly
923 from the REML fit.

924 C.5.2 Method B: Generalized Linear Mixed Effects (GLME)

925 A logistic mixed-effects model with a Bernoulli likelihood and logit link is fit via Laplace approxima-
926 tion to the marginal likelihood, with parameter estimation by penalized iteratively reweighted least
927 squares (PIRLS). This respects the binary nature of the data but estimates are on the logit scale; we
928 convert variance proportions by computing the share of total variance (including the logistic residual
929 variance $\pi^2/3$) attributable to each component. Both the Linear and Generalized Linear models were
930 estimated using `lme4` [Bates et al., 2015].

931 C.5.3 Method C: Bayesian Generalized Linear Mixed Effects

932 We fit the same logistic specification as Method B within a fully Bayesian framework using Hamil-
933 tonian Monte Carlo (HMC) via `brms`/Stan [Bürkner, 2021]. Weakly informative half-Student- t
934 priors are placed on all standard-deviation parameters. Four chains of 1,000 (Models 1–3) or 2,000
935 (Model 4) iterations are run with thinning factors of 2 or 5, respectively, and `adapt_delta` = 0.95
936 to reduce divergent transitions. Convergence is assessed by $\hat{R} < 1.05$ for all parameters. Posterior
937 means, medians, and 95% highest-density intervals (HDIs) of variance proportions are reported,
938 providing a full uncertainty quantification absent from frequentist methods.

939 C.5.4 Method D: Distance Components (DISCO)

940 We estimate a nonparametric variance decomposition using distance components [Rizzo and Székely,
941 2010, Székely and Rizzo, 2017] from the energy-statistics literature. DISCO decomposes total
942 dispersion—measured by pairwise Euclidean distances—into between- and within-group components
943 without distributional assumptions. For a single facet with K groups,

$$\mathcal{S}_{\text{total}} = \mathcal{S}_{\text{between}} + \mathcal{S}_{\text{within}}, \quad (14)$$

944 where $\mathcal{S}$ denotes the energy-based dispersion statistic. The G-coefficient analog is $E\rho_{\text{DISCO}}^2 =$
945 $\mathcal{S}_p/(\mathcal{S}_p + \mathcal{S}_{\text{within}})$. DISCO captures nonlinear relationships and is robust to the heavy skewness
946 observed in the Bayesian posteriors, particularly for facets with few levels (e.g., agents). However, it
947 does not guarantee a positive $\sigma_{\epsilon}^2/\mathcal{S}_{\text{global within}}$ term, so its variance shares sum to one across the total
948 explained dispersion.

949 C.6 Reliability Estimation and Projection

950 From each fitted model, we extract the proportion of total variance (or dispersion, for DISCO)
951 attributable to each facet. We additionally compute *projected* G-coefficients under hypothetical
952 increases in the number of items, using the Spearman–Brown-like formula from G-theory:

$$E\rho^2(n'_I) = \frac{\sigma_m^2}{\sigma_m^2 + \frac{1}{n'_I}(\sigma_{tm}^2 + \sigma_{\epsilon}^2)}, \quad (15)$$

953 where n'_I is the projected item count. For Model 4, analogous projections vary both item and
954 benchmark counts.

955 C.7 Reliability Projection Formula

956 For a $p \times I$ design (models crossed with items), the projected G-coefficient under n_I items is

$$E\rho^2(n_I) = \frac{\sigma_m^2}{\sigma_m^2 + \frac{\sigma_{mI}^2 + \sigma_{\epsilon}^2}{n_I}}, \quad (16)$$

957 where σ_{mI}^2 is the model $\times$ item interaction variance. As $n_I \rightarrow \infty$, $E\rho^2 \rightarrow \sigma_m^2/\sigma_m^2$, but only if facets
958 that do not diminish with more items (e.g., agent main effects, benchmark $\times$ model interactions) are
959 negligible. When such facets are present, the asymptote is

$$\lim_{n_I \rightarrow \infty} E\rho^2 = \frac{\sigma_m^2}{\sigma_m^2 + \sigma_{ma}^2 + \sigma_{mb}^2 + \dots} < 1, \quad (17)$$

960 producing the reliability ceiling documented in our Model 4 analysis.

961 C.8 Variance Proportion Tables

962 Full variance-proportion tables for all four models and all four estimation methods are provided in the
963 supplementary materials. Tables in the main text report Bayesian posterior summaries; the appendix
964 tables additionally include LME, GLME, and DISCO estimates to enable cross-method comparison.

965 C.9 Latent Space Estimations

966 Variance components in our GLMM decompositions live on the *latent* (logit) scale, so ρ^2 should
967 be interpreted as reliability of the linear predictor η (i.e., of differences in *log-odds*) rather than
968 as reliability of raw percent-correct. This is standard in binary-response measurement: the latent
969 scale is the scale on which additive random effects and G-theory D-study algebra apply, and it is the
970 scale on which “true score + error” is well-defined without probability-dependent heteroskedasticity.
971 Importantly, a given amount of latent variability implies different variability in percent-correct
972 depending on where a model sits on the sigmoid: near $p \approx 0.5$, small changes in η translate to
973 large changes in p , while near $p \approx 0$ or 1 they compress. For that reason, a latent-scale reliability

such as $\rho^2 \approx 0.45$ *does* correspond to meaningful rank instability on the probability scale for the mid-performing region typical of many AI-agent benchmarks (where $dp/d\eta = p(1-p)$ is largest), but its practical impact (if that were desired) should be assessed by mapping latent uncertainty through the inverse-logit.

Accordingly, in supplementary material, we report an additional observed-scale sensitivity analysis: for each posterior draw of variance components we (i) generate replicated item averages for each object under the fitted GLMM, (ii) transform to percent-correct $\bar{p}$ via logit^{-1} , and (iii) summarize rank variability (e.g., expected Kendall- τ or pairwise reversal probability) on the probability scale. This makes clear that our conclusions are not an artifact of a hard-to-interpret latent metric: the latent-scale ρ^2 is the mathematically coherent reliability target for binary data, and the associated posterior predictive mapping quantifies how it manifests as practically relevant leaderboard instability in percent-correct.

C.10 Identifiability of the pooled ecosystem decomposition (Decomposition 4)

Decomposition 4 pools all benchmarks to estimate variance components for models, agent harnesses, benchmark effects, and their interactions. Because the evaluation matrix is sparse and unbalanced, we argue identifiability based on (i) *which parameters are identified from connectivity* in the observed design, (ii) *which are only weakly identified* (hence wide posteriors), and (iii) why the observed “structural ceiling” in model-ranking reliability is not a generic artifact of hierarchical shrinkage.

C.10.1 Design graph and why connectivity matters

Let an observation be indexed by a tuple $x = (b, i, m, a)$: benchmark $b \in \mathcal{B}$, item $i \in \mathcal{I}_b$, model $m \in \mathcal{M}$, and agent harness $a \in \mathcal{A}$. Write the linear predictor of Decomposition 4 as

$$\begin{aligned} \eta_{bima} = & \beta_0 + u_b^{(B)} + u_{b:i}^{(BI)} + u_m^{(M)} + u_a^{(A)} + u_{b:i:m}^{(BIM)} + u_{b:i:a}^{(BIA)} \\ & + u_{m:a}^{(MA)} + u_{b:i:m:a}^{(BIMA)} + u_{b:m}^{(BM)} + u_{b:a}^{(BA)} + u_{b:m:a}^{(BMA)}, \end{aligned} \quad (18)$$

with $y_{bima} \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_{bima}))$ and independent Gaussian random effects (mean zero, component variances σ_k^2). A convenient way to reason about identifiability in sparse crossed random-effects models is via a *connectivity graph*. Define a bipartite graph

$$G_b = (\mathcal{M} \cup \mathcal{A}_b, E_b), \quad (m, a) \in E_b \Leftrightarrow \exists i \in \mathcal{I}_b : y_{bima} \text{ observed.}$$

Intuitively, if G_b has multiple disconnected components, then within benchmark b we cannot compare models across components because no agent harness provides a common measurement context. Decomposition 4 requires *cross-benchmark* connectivity as well. For that, define the global tripartite incidence structure induced by observed tuples (b, m, a) :

$$(m, a, b) \text{ is present} \Leftrightarrow \exists i \in \mathcal{I}_b : y_{bima} \text{ observed.}$$

Role of HAL-generalist. One harness, denoted a^* (HAL-generalist), is evaluated on *every* benchmark and appears in all benchmarks that have only two harnesses. This induces a star-like connectivity pattern: for each b , many models satisfy $(m, a^*) \in E_b$, and across benchmarks the same a^* provides a shared reference harness.

This matters because the decomposition includes both (i) *global harness effects* $u_a^{(A)}$ and (ii) *benchmark-conditioned harness effects* $u_{b:a}^{(BA)}$ as well as interactions with models. Without a harness present across benchmarks, $u_m^{(M)}$ and $u_{b:m}^{(BM)}$ (or $u_a^{(A)}$ and $u_{b:a}^{(BA)}$) can become only weakly separated: differences could be attributed either to model quality or to unobserved changes in harness composition.

C.10.2 Model identification

(i) Identifiability of latent effects vs identifiability of variance components. Random effects are latent variables. In GLMMs, individual latent effects (e.g., a particular $u_{b:m}^{(BM)}$) are only identifiable up to their contribution to η , and can be weakly identified when few observations touch that level. What we require for reliability analysis is stronger: *identifiability of variance components* σ_k^2 , since ρ^2 is a functional of $\{\sigma_k^2\}$ under a D-study.

1017 **(ii) Likelihood identifiability vs Bayesian propriety.** A Bayesian posterior can be proper even
 1018 when the likelihood is weakly identified, due to priors. Our goal is to argue that the key conclusion (a
 1019 ceiling on model-ranking reliability) is supported by the *data connectivity*, not created by priors. Con-
 1020 cretely: the data must support separating model variability from harness and benchmark variability,
 1021 at least along the subspace needed to upper bound achievable ρ^2 as item count grows.

1022 C.10.3 Anchoring and separating global vs benchmark-conditioned effects

1023 We assume the standard random-effects centering constraints implied by the model specification:

$$\mathbb{E}[u^{(k)}] = 0 \quad \text{for all random-effect families } k,$$

1024 which removes additive non-identifiability with the intercept. The remaining concern is *aliasing*
 1025 among crossed terms, e.g., whether $u_m^{(M)}$ can be traded off with $u_{m:a}^{(MA)}$ or $u_{b:m}^{(BM)}$ without changing η
 1026 on observed cells.

1027 The HAL-generalist harness provides a measurement anchor that breaks key aliasing directions.

1028 **Lemma C.2** (HAL anchoring identifies global model contrasts across benchmarks). *Assume:*

- 1029 1. **(Cross-benchmark anchor)** *There exists an agent harness a^* such that for every benchmark*
 1030 *$b \in \mathcal{B}$, at least two distinct models $m_1, m_2 \in \mathcal{M}$ are observed with a^* on some items in*
 1031 *benchmark b .*
- 1032 2. **(Within-benchmark item replication)** *For each observed (b, m, a^*) , scores are observed on*
 1033 *at least two items (so that item effects do not perfectly saturate the cell).*

1034 *Then for any pair of models (m_1, m_2) that both appear with a^* in at least one common benchmark,*
 1035 *the data identify their global contrast $u_{m_1}^{(M)} - u_{m_2}^{(M)}$ up to benchmark-conditioned residuals that*
 1036 *average out under item resampling; i.e., the contrast cannot be represented solely as a reallocation*
 1037 *into agent-only or benchmark-only terms without changing η on observed cells.*

1038 *Sketch.* Consider two observations in the same benchmark b and harness a^* but different models, and
 1039 average across items:

$$\bar{\eta}_{b,m,a^*} := \frac{1}{|\mathcal{I}_{b,m,a^*}|} \sum_{i \in \mathcal{I}_{b,m,a^*}} \eta_{bima^*}.$$

1040 Within the same (b, a^*) , terms $u_b^{(B)}$, $u_{a^*}^{(A)}$, and $u_{b:a^*}^{(BA)}$ cancel in the difference $\bar{\eta}_{b,m_1,a^*} - \bar{\eta}_{b,m_2,a^*}$.
 1041 Item-indexed terms shrink under averaging (and, in a D-study, under increasing item count). What
 1042 remains includes $u_{m_1}^{(M)} - u_{m_2}^{(M)}$ plus model-conditioned interactions. Because a^* is fixed across
 1043 both models, the model contrast cannot be fully absorbed into purely harness or benchmark main
 1044 effects. $\square$

1045 **Interpretation.** Lemma C.2 is not claiming perfect analytic identifiability of every interaction term.
 1046 Rather, it establishes that the observed design contains a repeated, shared harness that enables *global*
 1047 *model comparisons* to be learned from data, which is the minimum needed to interpret σ_M^2 and hence
 1048 model-ranking reliability.

1049 C.10.4 Observed ceiling would not be a hierarchical artifact

1050 A common misunderstanding when interpreting larger hierarchical models is that when one such
 1051 model is fit on sparse data, it will shrink everything and produce low reliability. In truth, shrinkage can
 1052 increase or decrease estimated σ_M^2 depending on how the data allocate variation among components.
 1053 This is more clearly decided by whether the model is forced by its structure to attribute large mass to
 1054 non-model components even when the data would support a high σ_M^2 .

1055 Any ceiling in Decomposition 4 would arise because it includes components that do *not* vanish
 1056 with more items, and these components are empirically non-negligible under multiple estimators
 1057 (LME, GLME, Bayesian, DISCO). *Even an infinite-item D-study cannot remove variability due to*
 1058 *benchmark-conditioned model and harness interactions.*

Proposition C.3 (Non-vanishing error terms impose an upper bound on ρ_{model}^2). *Consider the pooled model with object of measurement m and a decision rule that aggregates scores by averaging over n_i items within each benchmark and a fixed harness mixture (as in observed leaderboards). Let σ_M^2 be the global model variance and let σ_{nv}^2 be the sum of variance components that are not attenuated by increasing n_i under this design (e.g., benchmark $\times$ model, model $\times$ agent, benchmark $\times$ model $\times$ agent, and other terms not scaled by $1/n_i$ in the D-study). Then the D-study model-ranking reliability satisfies*

$$\limsup_{n_i \rightarrow \infty} \rho_{\text{model}}^2(n_i) \leq \frac{\sigma_M^2}{\sigma_M^2 + \sigma_{nv}^2}.$$

In particular, if $\sigma_{nv}^2 > 0$, reliability is structurally bounded away from 1 even with arbitrarily many items.

Sketch. In standard G-theory, averaging over n_i items attenuates components involving i by factors of $1/n_i$ (or $1/(n_i n_a)$ if also averaging over agents). Components that do not involve i (or that remain due to fixed harness mixtures and benchmark-conditioned shifts) remain. Reliability is signal over signal+error, hence the bound. $\square$

Not a result of a larger measurement model Proposition C.3 shows that the only way to avoid a ceiling is for the data to imply $\sigma_{nv}^2 \approx 0$ (or to change the decision design, e.g., average over many harnesses and benchmarks). In our estimates, non-item components such as benchmark effects and benchmark-conditioned interactions remain appreciable across estimation methods, which is evidence that the ceiling reflects measured ecosystem heterogeneity rather than prior-driven shrinkage.

C.10.5 Connectivity is sufficient for the present conclusions

The paper’s main conclusion is *not* that every variance component in Eq. equation 18 is precisely identified. Rather, it is that the ecosystem contains enough cross-context variability that model-ranking reliability cannot be made “high” by scaling item counts alone. For this, we need two claims:

Claim A: σ_M^2 is not spuriously driven to zero. Because a^* appears across all benchmarks, global model contrasts are repeatedly observed under a common harness, preventing the model effect from being purely absorbed into benchmark or harness terms. This guards against the pathological case where σ_M^2 is an artifactually small residual category.

Claim B: non-vanishing heterogeneity is data-supported. Benchmark-conditioned terms (e.g., $u_{b:m}^{(BM)}$, $u_{b:m:a}^{(BMA)}$) are supported by repeated observations within each benchmark across multiple models and at least one shared harness. In other words, the model is not inferring benchmark-conditioned interactions from a single cell; it is using within-benchmark comparisons and cross-benchmark anchoring via a^* .

We summarize these two claims as:

Theorem C.4 (Sufficient connectivity for bounding model-ranking reliability). *Assume (i) there exists a cross-benchmark anchor harness a^* observed in every benchmark, (ii) within each benchmark at least two models are evaluated under a^* on multiple items, and (iii) at least one additional harness appears in some benchmarks (so that model $\times$ agent variability is not degenerate). Then:*

1. *The global model variance σ_M^2 is estimable from repeated anchored contrasts and is not identifiable only through the prior.*
2. *If the data support $\sigma_{nv}^2 > 0$ (in the sense that posterior mass for these components is away from zero under weakly informative priors and is corroborated by nonparametric DISCO decomposition), then Proposition C.3 implies a structural upper bound on ρ_{model}^2 that persists as n_i grows.*

Discussion. Condition (i) is satisfied by HAL-generalist; condition (ii) is satisfied because each benchmark evaluates many models across many items under HAL-generalist; condition (iii) is satisfied because several benchmarks include additional harnesses (2–3 total), which is enough to reveal nontrivial $M \times A$ and benchmark-conditioned harness effects. Thus, while some high-order

terms remain weakly identified (wide posteriors), the *existence* of non-vanishing heterogeneity—and hence a ceiling on ρ_{model}^2 under item-scaling—is robust to modeling choices.

C.10.6 Practical takeaway for benchmark designers

To make model leaderboards stable, one must increase *connectivity*, not just sample size. The single most valuable design primitive is a *bridge harness* (like HAL-generalist) evaluated across benchmarks, plus enough additional harness diversity to estimate $M \times A$ variability. If a benchmark ecosystem lacks such anchors, pooled decompositions become prior-sensitive and any reliability claim becomes fragile. Here, the observed anchor connectivity ensures that the measured reliability ceiling reflects real cross-context variability in agentic evaluation rather than a hierarchical modeling artifact. This makes Decomposition 4 the strongest for interrogating the reliability of AI-agent benchmarking.

D Additional Plots

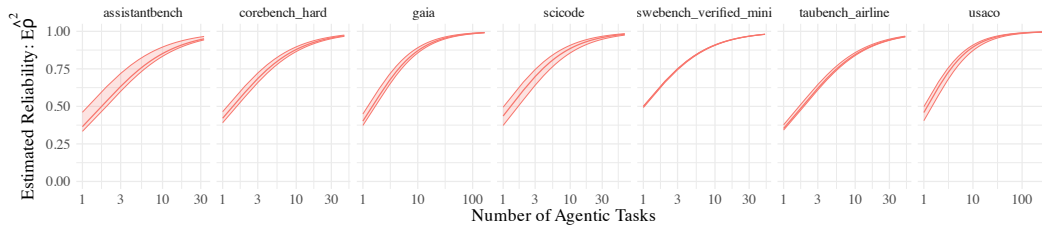


Figure 8: Naïve Estimated Rank Reliability. Variance components aggregated after inverse logit transform, thus reliabilities represent the observation space.

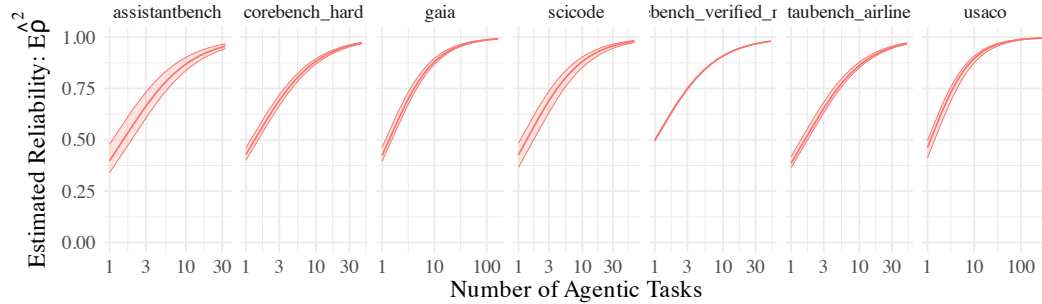


Figure 9: Estimated Reliability of Rankings across Agentic Benchmarks, measured by Model-Scaffold pairing. Variance components aggregated after inverse logit transform, thus reliabilities represent the observation space.

Table 4: Rank and Distance Correlations between Original Rankings and Best Linear Unbiased Predictor Rankings

benchmark	Kendall	dcor	Spearman
assistantbench	0.699	0.762	0.863
corebench_hard	0.717	0.783	0.880
gaia	0.434	0.449	0.637
scicode	0.757	0.910	0.911
swebench_verified_mini	0.694	0.728	0.863
taubench_airline	0.380	0.154	0.488
usaco	0.779	0.646	0.915

For most benchmarks we see that general rank order (Spearman) and distribution are preserved (dCor) Rank and Distance Correlations between Original Rankings and Best Linear Unbiased Predictor Rankings

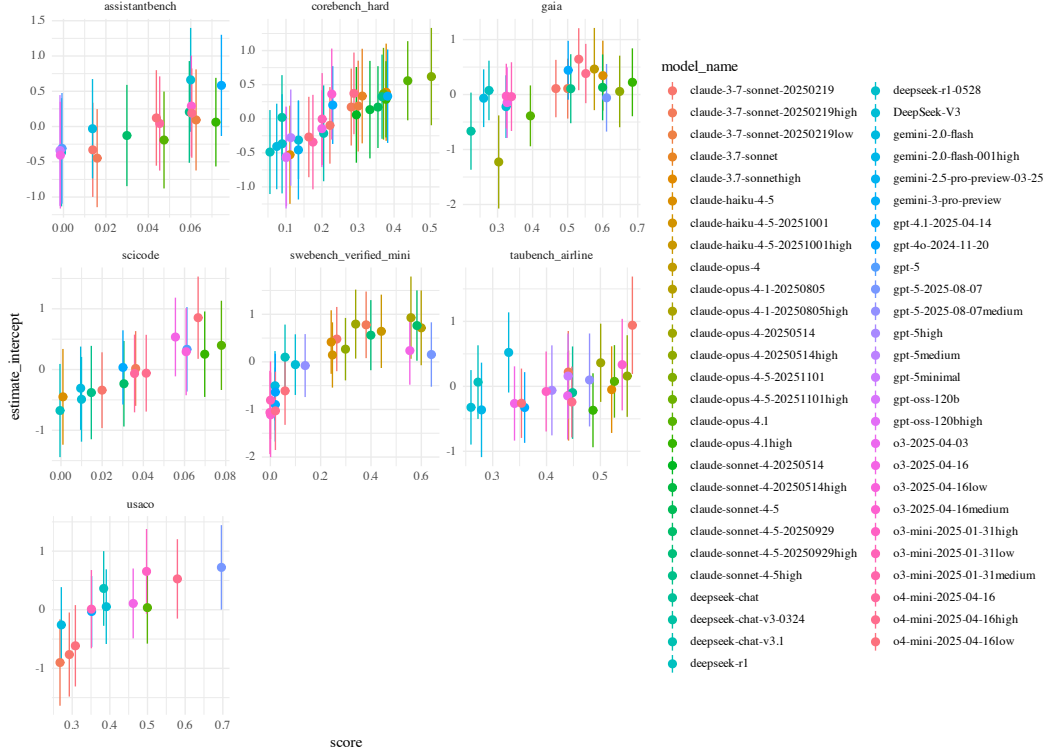


Figure 10: Published scores and estimated scores with standard errors estimated from the Best Linear Unbiased Predictors from the posterior of Eq. 3

E Method Comparison

E.1 Variance shares and reliability on the latent scale

For GLMM/Bayesian GLMM, we report variance shares as proportions of total latent variance:

$$\pi_k = \frac{\sigma_k^2}{\sum_j \sigma_j^2},$$

where the sum runs over included random effects and (optionally) a latent residual term. For Bayesian fits, π_k is computed per posterior draw, yielding credible intervals.

D-study scaling. When an interaction term involves a sampled facet, its contribution to the variance of an averaged score shrinks with the number of sampled levels. For example, with items averaged (n_i items), an $I \times M$ component contributes σ_{IM}^2/n_i .

E.2 Distance components (DISCO) reliability

DISCO decomposes dispersion using pairwise distances rather than squared deviations, improving robustness under non-normality and heavy-tailed random effects. For an object grouping g (e.g., model or model-scaffold pair), DISCO yields between-group and within-group dispersion terms (T_g, T_{within}), from which we define

$$E\rho^2 = \frac{T_g}{T_g + T_{\text{within}}}.$$

We use DISCO primarily as a sensitivity analysis to validate that conclusions (dominant facets; low model-ranking reliability) are not artifacts of Gaussian random-effect assumptions.

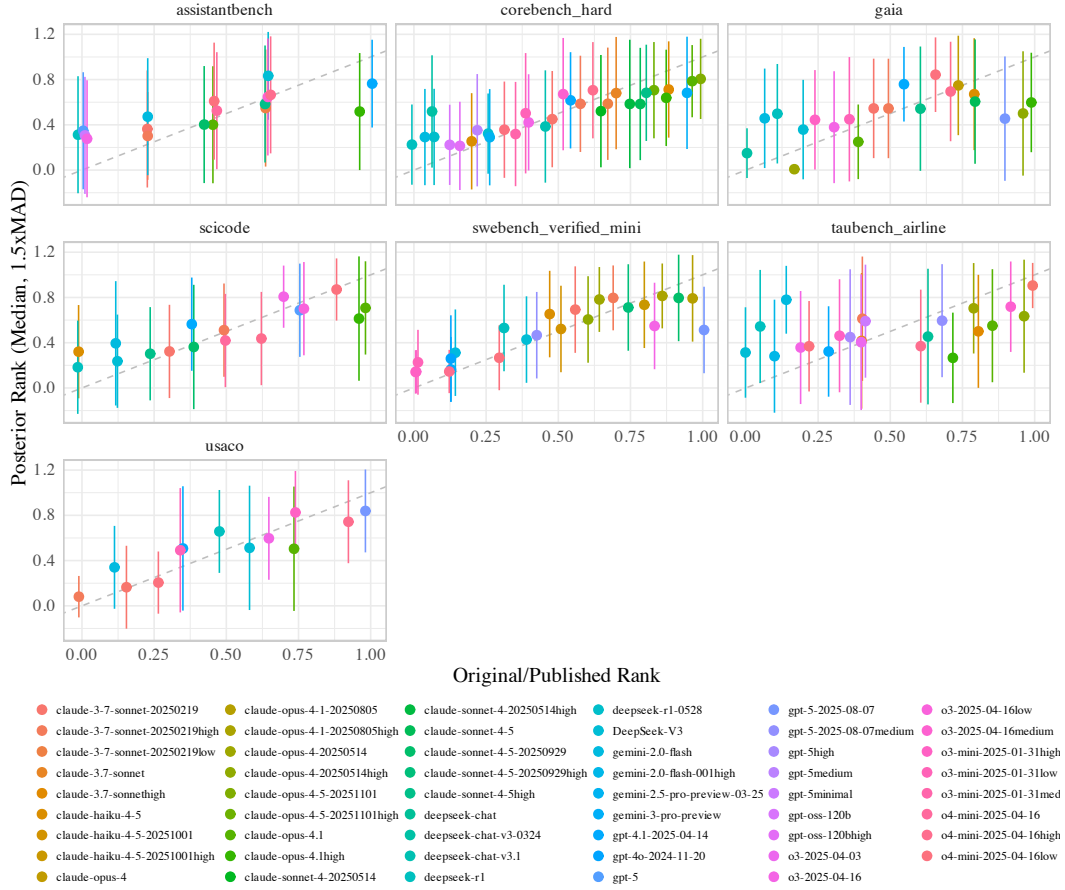


Figure 11: Published rankings and draw-wise posterior median rankings of BLUPs with 90% CIs

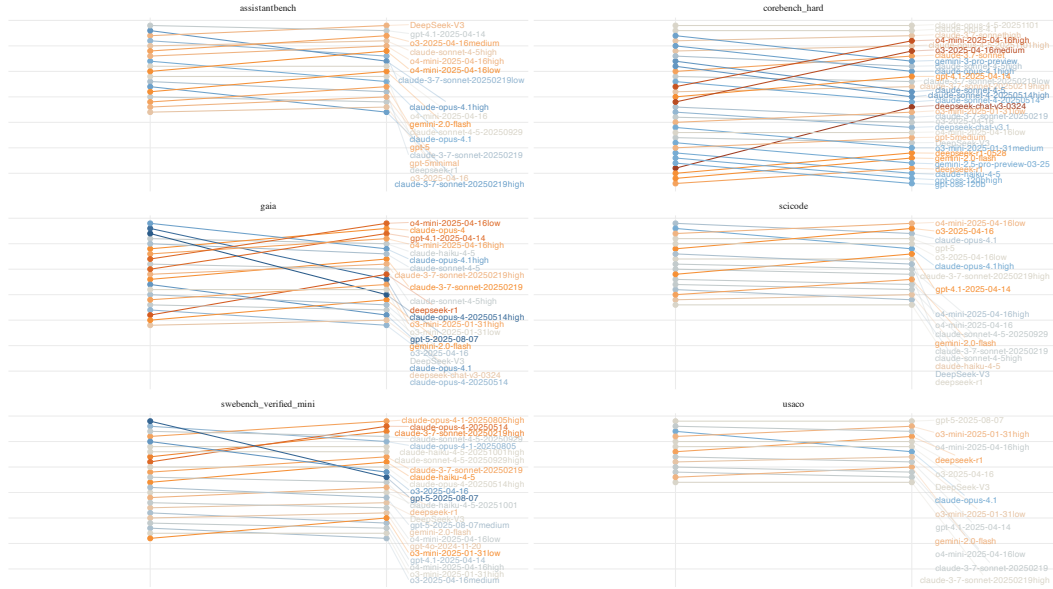


Figure 12: Rank order changes by benchmark based on comparison with Best Linear Unbiased Predictors (BLUPs) from full variance decomposition model of Eq. 3. Taubench Airline can be found in the main body Figure 5

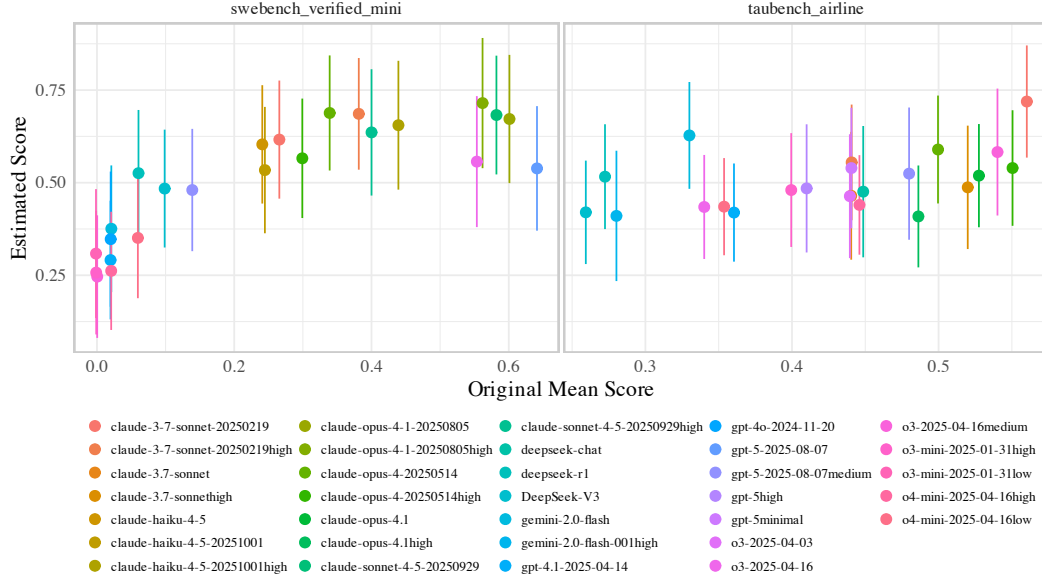


Figure 13: Published scores and estimated scores with standard errors estimated from the Best Linear Unbiased Predictors from the posterior of Eq. 3. Estimates for all benchmarks are in Figure 10. Variance components aggregated after inverse logit transform, thus reliabilities represent the observation space.

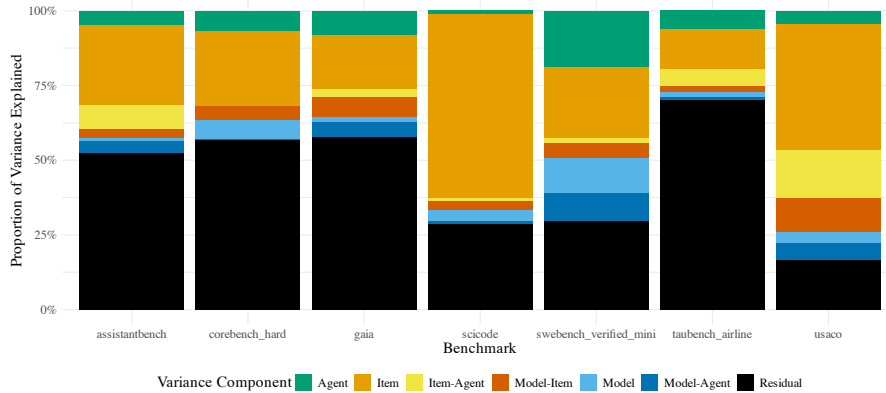


Figure 14: Proportion of Variance Explained in the posterior distributions from Eq. 2.

E.3 Why estimation method choice changes conclusions—and what to do about it

The four estimation methods are not interchangeable, and their disagreements are informative.

A notable outcome is that variance shares differ across LME, GLME, Bayesian GLMM, and DISCO, especially under sparse observations:

- **LME** tends to allocate a large portion to residual σ^2 , which can understate structured interactions when the link is misspecified for Bernoulli data.
- **GLME** can collapse some components toward zero (boundary estimates) in unbalanced designs, yielding deceptively “clean” decompositions.
- **Bayesian GLMM** exposes skewness and large uncertainty, which is appropriate when the design under-identifies components.
- **DISCO** can attribute more to interaction-like dispersion without parametric assumptions, often aligning with the intuition that agentic pipelines create nonlinear, heteroskedastic effects.

1145 **Practical guidance.**

- 1146 1. Use **Bayesian** or **nonparametric** methods to communicate uncertainty when data are sparse;
1147 prefer point-estimate GLME/LME only when coverage is dense.
1148 2. Triangulate: when all four methods agree on the *ranking* of dominant facets (e.g., item
1149 vs interactions vs residual), recommendations are robust; when they disagree, the correct
1150 conclusion is that the benchmark is under-instrumented for that inference.

1151 **Linear vs. generalized models.** The LME estimates consistently attribute the largest share of
1152 variance to the residual (45–85%), with correspondingly compressed named components. The GLME
1153 and Bayesian estimates, by contrast, allocate substantially more variance to items and models. This
1154 divergence is expected: the Gaussian identity-link model treats binary $\{0, 1\}$ responses as continuous,
1155 and its residual absorbs the Bernoulli variance floor ($p(1 - p)$) that the logit-link models separate out
1156 via the distributional assumption. The practical implication is that *linear G-theory estimates applied*
1157 *to pass/fail benchmarks systematically underestimate the proportion of variance attributable to*
1158 *named facets and overestimate residual noise*, leading to artificially compressed—but not necessarily
1159 more conservative—reliability estimates.

1160 **Bayesian vs. frequentist GLME.** The Bayesian and GLME estimates agree in broad strokes but
1161 diverge in two instructive ways. First, the Bayesian posteriors for agent variance are markedly
1162 right-skewed, with posterior means $2\text{--}5\times$ larger than posterior medians (e.g., SWE-bench agent
1163 mean = 0.27, median = 0.19). The GLME point estimate, which approximates the posterior mode,
1164 misses this tail mass and underestimates the expected contribution of scaffold variance. Second, the
1165 credible intervals from the Bayesian analysis expose the *decision-relevant uncertainty*: for several
1166 benchmarks, the 95% HDI for model variance includes zero, meaning the data are consistent with no
1167 true model differentiation at all. This uncertainty is invisible in frequentist analyses.

1168 **Nonparametric DISCO.** The DISCO estimates depart most dramatically from the parametric
1169 methods on the interaction terms. DISCO attributes 40–57% of total dispersion to item $\times$ model
1170 interactions across benchmarks, compared to 1–11% from the Bayesian estimates. This divergence
1171 likely reflects two mechanisms: (i) DISCO’s sensitivity to nonlinear dependencies that the additive
1172 random-effects models cannot capture, and (ii) the absence of a separate residual term in DISCO,
1173 which forces unexplained variation into the named interactions. The practical upshot is that DISCO
1174 serves as a useful *upper bound* on interaction effects and a reminder that the additive decomposition
1175 assumed by mixed-effects models may understate the complexity of the model $\times$ item relationship.

1176 **E.4 Full Estimation Tables for Proportion of Latent Variation Explained**

Table 5: Proportion of variation explained Decomposition 1

benchmark	facet	nonp	lme	glme	bayes
scicode	model	0.076	0.012	0.050	0.047
scicode	item	0.924	0.254	0.784	0.601
gaia	model	0.313	0.078	0.104	0.055
gaia	item	0.687	0.231	0.342	0.155
taubench_airline	model	0.268	0.025	0.033	0.016
taubench_airline	item	0.732	0.196	0.300	0.125
swebench_verified_mini	model	0.578	0.234	0.440	0.378
swebench_verified_mini	item	0.422	0.196	0.323	0.261
usaco	model	0.110	0.067	0.100	0.088
usaco	item	0.890	0.479	0.663	0.446
assistantbench	model	0.171	0.006	0.037	0.027
assistantbench	item	0.829	0.149	0.508	0.286
corebench_hard	model	0.293	0.073	0.121	0.067
corebench_hard	item	0.707	0.242	0.447	0.245
scicode	sigma		0.734	0.166	0.361
gaia	sigma		0.692	0.554	0.783
taubench_airline	sigma		0.780	0.667	0.841
swebench_verified_mini	sigma		0.570	0.237	0.597
usaco	sigma		0.454	0.237	0.492
assistantbench	sigma		0.845	0.455	0.678
corebench_hard	sigma		0.684	0.432	0.691

Table 6: Proportion of variation explained Decomposition 2

benchmark	facet	nonp	lme	glme	bayes
scicode	model	0.076	0.012	0.047	0.040
scicode	item	0.924	0.254	0.787	0.602
gaia	model	0.313	0.078	0.144	0.076
gaia	item	0.687	0.231	0.360	0.171
taubench_airline	model	0.268	0.025	0.092	0.041
taubench_airline	item	0.732	0.196	0.304	0.136
swebench_verified_mini	model	0.578	0.234	0.436	0.390
swebench_verified_mini	item	0.422	0.196	0.342	0.285
usaco	model	0.110	0.067	0.104	0.091
usaco	item	0.890	0.479	0.667	0.456
assistantbench	model	0.171	0.006	0.072	0.047
assistantbench	item	0.829	0.149	0.501	0.302
corebench_hard	model	0.293	0.073	0.134	0.072
corebench_hard	item	0.707	0.242	0.450	0.250
scicode	sigma		0.734	0.166	0.363
gaia	sigma		0.692	0.496	0.757
taubench_airline	sigma		0.780	0.605	0.809
swebench_verified_mini	sigma		0.570	0.222	0.574
usaco	sigma		0.454	0.230	0.482
assistantbench	sigma		0.845	0.427	0.649
corebench_hard	sigma		0.684	0.417	0.684

Table 7: Proportion of variation explained Decomposition 3

benchmark	facet	nonp	lme	glme	bayes
scicode	item	0.209	0.235	0.766	0.590
scicode	agent	0.001	0.001	0.004	0.055
scicode	model	0.012	0.010	0.053	0.046
scicode	item_agent	0.252	0.037	0.012	0.015
scicode	item_model	0.509	0.130	0.012	0.029
scicode	model_agent	0.017	0.002	0.003	0.011
gaia	item	0.169	0.210	0.342	0.182
gaia	agent	0.017	0.041	0.029	0.172
gaia	model	0.046	0.030	0.040	0.024
gaia	item_agent	0.209	0.036	0.035	0.024
gaia	item_model	0.481	0.097	0.041	0.068
gaia	model_agent	0.077	0.059	0.083	0.053
taubench_airline	item	0.161	0.166	0.277	0.134
taubench_airline	agent	0.021	0.043	0.044	0.123
taubench_airline	model	0.025	0.027	0.039	0.020
taubench_airline	item_agent	0.250	0.086	0.107	0.057
taubench_airline	item_model	0.484	0.035	0.000	0.021
taubench_airline	model_agent	0.059	0.012	0.016	0.011
swebench_verified_mini	item	0.092	0.153	0.349	0.261
swebench_verified_mini	agent	0.073	0.246	0.103	0.329
swebench_verified_mini	model	0.119	0.056	0.272	0.136
swebench_verified_mini	item_agent	0.190	0.068	0.024	0.023
swebench_verified_mini	item_model	0.399	0.089	0.009	0.057
swebench_verified_mini	model_agent	0.126	0.057	0.042	0.119
usaco	item	0.239	0.257	0.551	0.420
usaco	agent	0.003	0.072	0.000	0.102
usaco	model	0.027	0.062	0.000	0.048
usaco	item_agent	0.262	0.200	0.130	0.163
usaco	item_model	0.440	0.184	0.000	0.112
usaco	model_agent	0.030	0.000	0.103	0.060
assistantbench	item	0.156	0.116	0.437	0.254
assistantbench	agent	0.002	0.000	0.000	0.217
assistantbench	model	0.017	0.000	0.000	0.020
assistantbench	item_agent	0.219	0.083	0.107	0.090
assistantbench	item_model	0.574	0.047	0.000	0.044
assistantbench	model_agent	0.032	0.013	0.069	0.043
corebench_hard	item	0.172	0.239	0.446	0.251
corebench_hard	agent	0.007	0.016	0.019	0.175
corebench_hard	model	0.061	0.064	0.114	0.065
corebench_hard	item_agent	0.187	0.006	0.000	0.002
corebench_hard	item_model	0.502	0.111	0.036	0.046
corebench_hard	model_agent	0.071	0.009	0.007	0.006
scicode	sigma		0.584	0.150	0.283
gaia	sigma		0.527	0.431	0.571
taubench_airline	sigma		0.631	0.518	0.684
swebench_verified_mini	sigma		0.331	0.202	0.319
usaco	sigma		0.225	0.216	0.166
assistantbench	sigma		0.741	0.386	0.482
corebench_hard	sigma		0.555	0.378	0.556

Table 8: Proportion of variation explained Decomposition 4

facet	lme	glme	gbayes	nonp
agent	0.007	0.008	0.034	0.020
benchmark	0.097	0.287	0.268	0.030
benchmark_agent	0.035	0.032	0.026	0.037
benchmark_item	0.262	0.345	0.257	0.110
benchmark_item_agent	0.041	0.026	0.024	0.128
benchmark_item_model	0.063	0.011	0.033	0.229
benchmark_model	0.013	0.024	0.015	0.049
benchmark_model_agent	0.011	0.007	0.010	0.056
model	0.030	0.037	0.029	0.020
model_agent	0.008	0.013	0.009	0.043
benchmark_item_model_agent + sigma	0.433	0.210	0.304	0.278